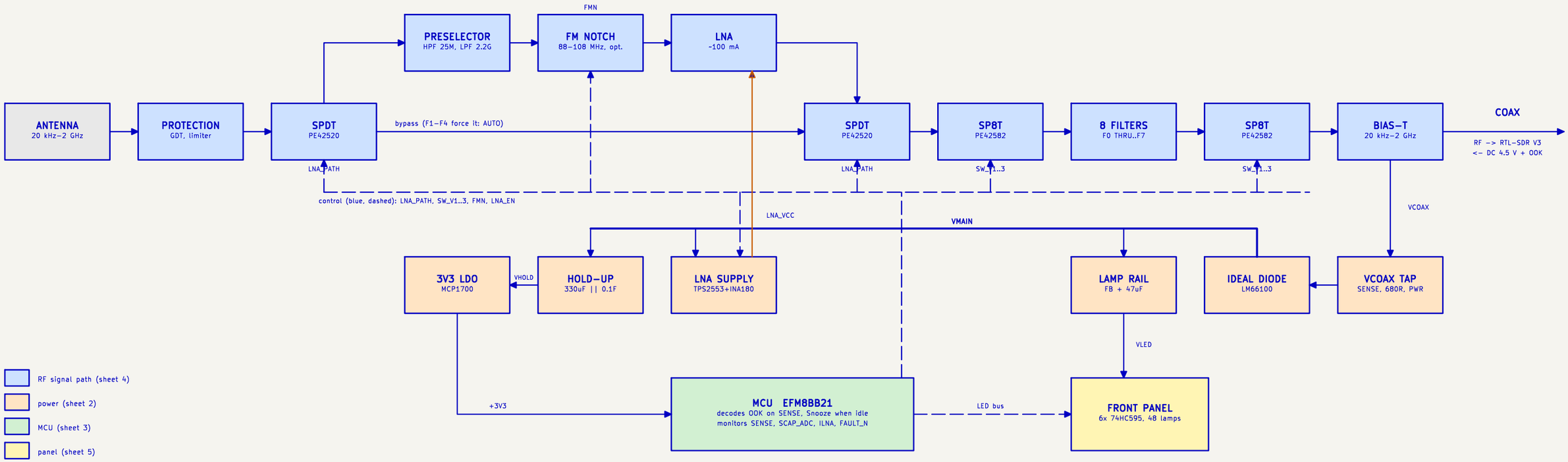
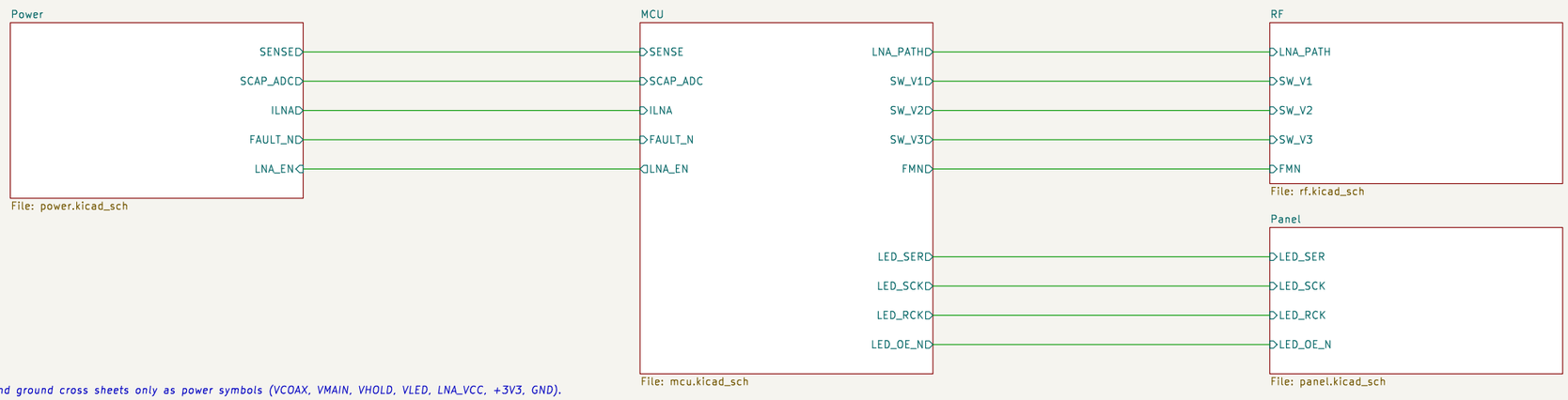
Coaxolotl — block diagram



Sheets

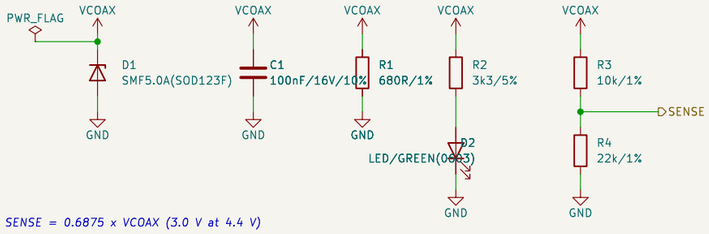


Power and ground cross sheets only as power symbols (VCOAX, VMAIN, VHOLD, VLED, LNA_VCC, +3V3, GND).
Signals cross sheets as hierarchical labels through the sheet pins above. Firmware pin map: firmware/src/hal_efm8bb2.c

Generated by hardware/kicad/tools/build.py — edit the Python, not this file
Bias-T powered, OOK switched LNA + 8-way filter bank
z2labs

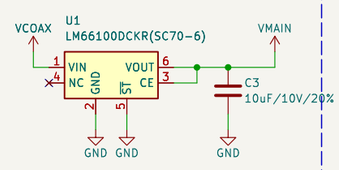
Sheet: /		
File: coaxolotl.kicad_sch		
Title: Coaxolotl		
Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5

VCOAX tap



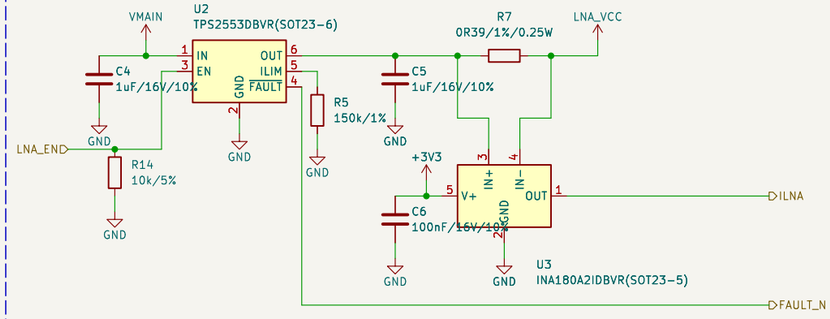
SENSE = 0.6875 x VCOAX (3.0 V at 4.4 V)
t_fall = 1.1 x R_eff x C_total: R1 sets the OOK falling edge

Ideal diode



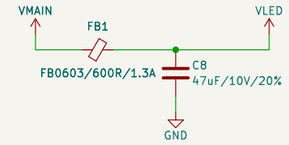
CE = VOUT: comparator blocks
reverse current in OOK gaps

LNA supply + current monitor



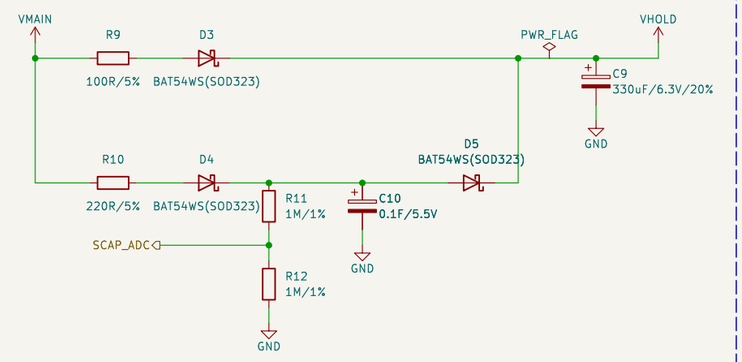
Ilim = f(R5), TPS2553 Eq.1: VERIFY IOS(min) > 130 mA
ILNA = 0.39R x 50 V/V = 19.5 mV/mA (100 mA -> 1.95 V); Kelvin taps at R7 pads

Lamp rail



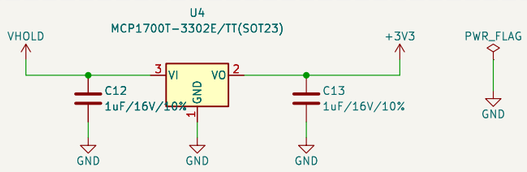
48 lamps x 1 mA max on VMAIN:
dark in OOK gaps, no hold-up drain

Hold-up (diode-OR)



fast path R9/D3: VHOLD up in ~30 ms (100R x 330uF) -> MCU boots
slow path R10/D4: tau ~22 s (220R x 0.1F); ride-through ~4 min at 0.6 mA
SCAP_ADC = VSCAP / 2 (HOLD lamp at >= 3.6 V)

3V3 regulator



MCP1700: Iq 1.6 uA, dropout 178 mV;
VO needs >= 1 uF ceramic

Generated by hardware/kicad/tools/build.py — edit the Python, not this file
VCOAX tap, ideal diode, LNA supply switch + current monitor, lamp rail, hold-up, 3V3

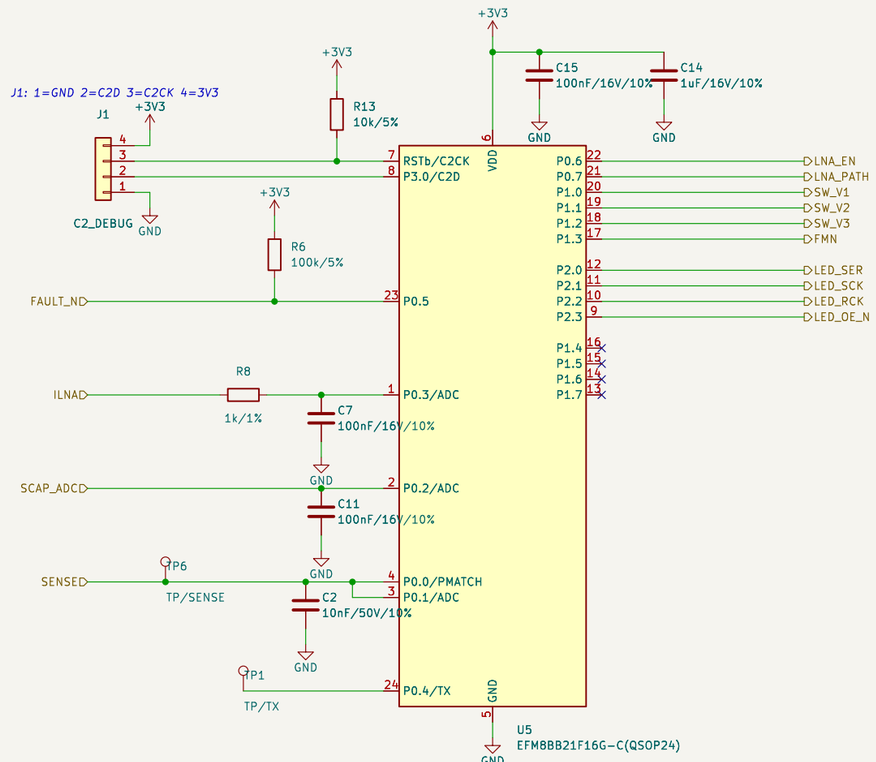
z2labs

Sheet: /Power/
File: power.kicad_sch

Title: Power

Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/5

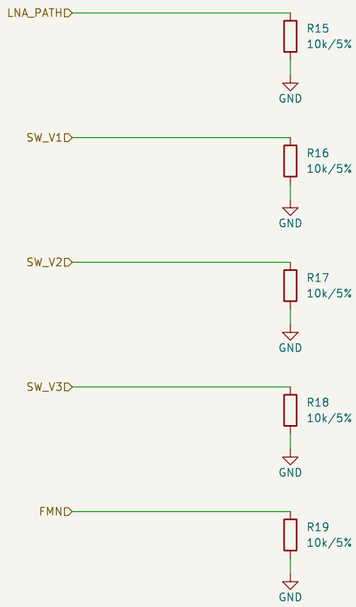
MCU



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EFM8BB21F16G, C2 debug, ADC input conditioning
z2labs

Sheet: /MCU/		
File: mcu.kicad_sch		
Title: MCU		
Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5

Switch control inputs



Pull-downs at the receiving switches: bypass, F0 = THRU, FM notch out while the MCU is in reset.

NEXT REVIEW: antenna protection, LNA/bypass PE42520, preselector, FM notch, 2x PE42582, 8 filters, broadband bias-T (drives VCOAX).

Generated by hardware/kicad/tools/build.py — edit the Python, not this file
RF chain — placeholder: control inputs only (next review)

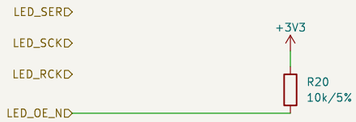
z2labs

Sheet: /RF/
File: rf.kicad_sch

Title: RF

Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/5

Lamp chain inputs



NEXT REVIEW: 6x 74HC595 (sinking), 48 lamps on VLED, 2.2k arrays.

Generated by hardware/kicad/tools/build.py — edit the Python, not this file
48-lamp front panel — placeholder (next review)

z2labs

Sheet: /Panel/		
File: panel.kicad_sch		
Title: Panel		
Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5